

$$\begin{aligned}
 & \begin{pmatrix} -53/7 & -58/7 & 0 \end{pmatrix} \\
 2) & \begin{pmatrix} 64/7 & 74/7 & 3 \\ -51/7 & -57/7 & 0 \\ -53/7 & -58/7 & 0 \end{pmatrix} \cdot \begin{pmatrix} -16/9 & 8/9 & -1/9 \\ 14/9 & -7/9 & 2/9 \\ -1/9 & 2/9 & -1/9 \end{pmatrix} = \\
 = & \begin{pmatrix} \frac{64}{7} \cdot (-\frac{16}{9}) + \frac{74}{7} \cdot \frac{14}{9} + 3 \cdot (-\frac{1}{9}) & \frac{64}{7} \cdot \frac{8}{9} + \frac{74}{7} \cdot (-\frac{7}{9}) + 3 \cdot (-\frac{1}{9}) & \frac{64}{7} \cdot (-\frac{1}{9}) + \frac{74}{7} \cdot \frac{2}{9} + 3 \cdot (-\frac{1}{9}) \\ -\frac{51}{7} \cdot (-\frac{16}{9}) + (-\frac{57}{7}) \cdot \frac{14}{9} + 0 \cdot (-\frac{1}{9}) & -\frac{51}{7} \cdot \frac{8}{9} + (-\frac{57}{7}) \cdot (-\frac{7}{9}) + 0 \cdot (-\frac{1}{9}) & -\frac{51}{7} \cdot (-\frac{1}{9}) + (-\frac{57}{7}) \cdot \frac{2}{9} + 0 \cdot (-\frac{1}{9}) \\ -\frac{53}{7} \cdot (-\frac{16}{9}) + (-\frac{58}{7}) \cdot \frac{14}{9} + 0 \cdot (-\frac{1}{9}) & -\frac{53}{7} \cdot \frac{8}{9} + (-\frac{58}{7}) \cdot (-\frac{7}{9}) + 0 \cdot (-\frac{1}{9}) & -\frac{53}{7} \cdot (-\frac{1}{9}) + (-\frac{58}{7}) \cdot \frac{2}{9} + 0 \cdot (-\frac{1}{9}) \end{pmatrix} \\
 & \begin{pmatrix} \frac{64}{7} \cdot (-\frac{1}{9}) + \frac{74}{7} \cdot \frac{2}{9} + 3 \cdot (-\frac{1}{9}) \\ -\frac{51}{7} \cdot (-\frac{1}{9}) + (-\frac{57}{7}) \cdot \frac{2}{9} + 0 \cdot (-\frac{1}{9}) \\ -\frac{53}{7} \cdot (-\frac{1}{9}) + (-\frac{58}{7}) \cdot \frac{2}{9} + 0 \cdot (-\frac{1}{9}) \end{pmatrix} = \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix}
 \end{aligned}$$

Линейная работа

с ЧАУ

2.1 32

$$\begin{cases} x_1 - x_2 = 1 \\ 2x_1 - 2x_2 = 5 \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 2 & -2 & 5 \end{array} \right) \xrightarrow{\Pi - 2I} \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 0 & 3 \end{array} \right)$$

$$r(A) \neq r(A|B) \Rightarrow \text{сист. несовм.}$$

Отмет: сист. несовм. нет решений

2. 1.33

$$\begin{cases} 3x + 2y = 5 \\ 6x + 4y = 10 \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 3 & 2 & 5 \\ 6 & 4 & 10 \end{array} \right) \xrightarrow{\Pi - 2I} \left(\begin{array}{cc|c} 3 & 2 & 5 \\ 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{сист. совм.}$$

$$n=2 > r \Rightarrow \text{сист. неопр.}$$

$$|\alpha_{11}|=3 \Rightarrow x_1 \text{ зл. перемен.}$$

y - св. перемен.

$$3x + 2y = 5$$

$$3x = 5 - 2y$$

$$x = \frac{5 - 2y}{3}$$

$$y = t, \text{ тогда } x = \frac{5 - 2t}{3}$$

$$O.P. = \left(\frac{5 - 2t}{3}; t \right)$$

$$\text{при } t=0 \text{ } O.P. \left(\frac{5}{3}; 0 \right)$$

$$\text{Отмет: (сист. совм., неопр.; } O.P. = \left(\frac{5 - 2t}{3}; t \right)$$

$$O.P. \left(\frac{5}{3}; 0 \right)$$

2. 1.34

$$\begin{cases} x_1 + 2x_2 = 3 \\ -2x_1 + 3x_2 = 0 \\ -2x_1 - 4x_2 = 1 \end{cases}$$

$$(A|B) \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 0 \\ -2 & -4 & 1 \end{pmatrix} \xrightarrow{\substack{II+2I \\ III+2I}} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 6 \\ 0 & 0 & 7 \end{pmatrix}$$

Сист. линейн

Ответ: система линейн, нет решений

2.136.

$$\begin{cases} 3x - y = -5 \\ 2x + 3y = 4 \\ -x + \frac{1}{3}y = \frac{5}{3} \\ x + 1.5y = 2 \end{cases}$$

$$(A|B) = \begin{pmatrix} 3 & -1 & -5 \\ 2 & 3 & 4 \\ -1 & \frac{1}{3} & \frac{5}{3} \\ x & 1.5 & 2 \end{pmatrix} \xrightarrow{\substack{3II-2I \\ III+\frac{1}{3}I \\ IV-\frac{1}{3}I}} \sim$$

$$\sim \begin{pmatrix} 3 & -1 & -5 \\ 0 & 11 & 22 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$r(A) = r(A|B) = 2$ - сист. совп.
 $n \neq 2 = r$ - сист. неспр.

$$\begin{cases} 3x - y = -5 \\ 11y = 22 \end{cases}$$

$$\begin{cases} 3x - y = -5 \\ y = 2 \end{cases}$$

$$\begin{cases} x = -1 \\ y = 2 \end{cases}$$

У.Р. и О.Р. $(-1; 2)$

Ответ: сист. совп., неспр. У.Р. и О.Р. $(-1; 2)$

2.1.37.

$$\begin{cases} 3x + 4y + 2z = 8 \\ 2x - 4y - 3z = -1 \\ x + 5y + z = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 3 & 4 & 2 & 8 \\ 2 & -4 & -3 & -1 \\ 1 & 5 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 3 & 4 & 2 & 8 \\ 0 & -20 & -13 & -19 \\ 0 & 11 & 1 & -8 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} 3 & 4 & 2 & 8 \\ 0 & -20 & -13 & -19 \\ 0 & 0 & -129 & -369 \end{array} \right)$$

$$r(A) = r(A|B) = \text{C.C.}$$

$$r = n = 3 = \text{C. dim.}$$

$$\begin{cases} 3x + 4y + 2z = 8 \\ -20y - 13z = -19 \\ -129z = -369 \end{cases}$$

$$\begin{cases} x = 2 \\ y = -1 \\ z = 3 \end{cases} \quad \text{U.P. u O.P.}(2, -1, 3)$$

Answer: uem. dim. cofu, U.P. u O.P.(2, -1, 3)

2.1.39

$$\begin{cases} 2x - y + z = 0 \\ 3x + 4y - 2z = 0 \\ 3x - 2y + 4z = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 3 & 4 & -2 & 0 \\ 3 & -2 & 4 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 0 & 11 & -4 & 0 \\ 0 & -2 & 4 & 0 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 0 & 11 & -4 & 0 \\ 0 & -1 & 11 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 0 & 11 & -4 & 0 \\ 0 & 0 & 120 & 0 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 0 & 11 & -4 & 0 \\ 0 & 0 & 120 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 3 - CC$$

$$n = 3 = r \Rightarrow C.O.$$

$$\begin{cases} 2x - y - 2 = 0 \\ 11y - 2 = 0 \\ 120 = 0 \end{cases} \begin{cases} x = 0 \\ y = 0 \\ z = 0 \end{cases}$$

$$OP \text{ u } ZP(0;0;0)$$

Проблем: како да се реши $OP(0;0;0)$

2.1.41

$$\begin{cases} 3x_1 + 2x_2 + x_3 = 5 \\ 2x_1 + 3x_2 + x_3 = 1 \\ 2x_1 + 5x_2 + 3x_3 = 11 \\ 30x_1 + 4x_2 - x_3 = -5 \end{cases} \quad (A|B) = \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 2 & 3 & 1 & 1 \\ 2 & 1 & 3 & 11 \\ 3 & 4 & -1 & -5 \end{array} \right) \begin{array}{l} \\ 3II - 2I \\ 3III - 2I \\ IV - I \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 0 & 5 & 1 & -7 \\ 0 & -1 & 7 & 23 \\ 0 & 2 & -2 & -10 \end{array} \right) \begin{array}{l} \\ \\ +III \cdot 5 + II \\ 5IV - 2II \end{array} \sim \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 0 & 5 & 1 & -7 \\ 0 & 0 & 36 & 108 \\ 0 & 0 & -12 & -36 \end{array} \right) \begin{array}{l} \\ \\ \\ 3IV + III \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 0 & 5 & 1 & -7 \\ 0 & 0 & 36 & 108 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 3 - CC$$

$$n = 3 = r - \text{comp.}$$

$$\begin{cases} 3x_1 + 3x_2 + x_3 = 5 \\ 5x_2 + x_3 = 7 \\ 36x_3 = 108 \end{cases} \quad \begin{cases} x_1 = 2 \\ x_2 = -2 \\ x_3 = 3 \end{cases}$$

OPK $\varphi P(2; -2; 3)$

Omletm: luvn. covu, onp. $(2; -2; 3)$

2.1.42

$$\begin{cases} 2\sqrt{5}x_1 - x_2 + \sqrt{5}x_3 = 1 \\ 10x_1 - \sqrt{5}x_2 + 5x_3 = \sqrt{5} \\ -2x_1 + \frac{\sqrt{5}}{5}x_2 - x_3 = -\frac{1}{\sqrt{5}} \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 2\sqrt{5} & -1 & \sqrt{5} & 1 \\ 10 & -\sqrt{5} & 5 & \sqrt{5} \\ -2 & \frac{\sqrt{5}}{5} & -1 & -\frac{1}{\sqrt{5}} \end{array} \right) \xrightarrow{\text{II} - \sqrt{5} \cdot \text{I}} \left(\begin{array}{ccc|c} 2\sqrt{5} & -1 & \sqrt{5} & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 - \text{c.c.}$$

$n=3 > r$ - luvn. ne onp.

$$|a_{11}| = 2\sqrt{5} \neq 0$$

I - 2x. luvn.

x_1, x_3 - lb. luvn.

$$2\sqrt{5}x_1 - x_2 + \sqrt{5}x_3 = 1$$

$$x_1 = \frac{x_2 - \sqrt{5}x_3 + 1}{2\sqrt{5}}$$

$$x_2 = t_1$$

$$x_3 = t_2$$

$$\Rightarrow \text{O.P.} \left(\frac{t_1 - \sqrt{5}t_2 + 1}{2\sqrt{5}}; t_1; t_2 \right)$$

$$\text{npu } t_1 = t_2 = 0 \text{ } \varphi \text{ } \mathcal{P} \left(\frac{1}{2\sqrt{5}}; 0; 0 \right)$$

Проблем: да се намери ОП ($t_1, 5t_2 + 1$)
 $t_1, t_1, 2, 9 \mid 1, 0, 0$
 35

2.1.43

$$\begin{cases} 3x_1 + 4x_2 + x_3 + 2x_4 = 3 \\ 6x_1 + 8x_2 + 2x_3 + 5x_4 = 4 \\ 9x_1 + 12x_2 + 3x_3 + 10x_4 = 13 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 6 & 8 & 2 & 5 & 4 \\ 9 & 12 & 3 & 10 & 13 \end{array} \right) \sim \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 4 & 4 \end{array} \right) \begin{array}{l} \\ II - 2I \\ III - 3I \end{array}$$

$$\sim \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} r(A) = r(A|B) = 2 \\ n > r = 3 \text{ неопр.} \end{array}$$

$4 - 2 = 2$ сл. неопр.

$$\begin{vmatrix} a_{13} & a_{14} \\ a_{23} & a_{24} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \neq 0$$

x_3, x_4 - сл. неопр.

x_1, x_2 - об. неопр.

$$\begin{cases} 3x_1 + 4x_2 + x_3 + 2x_4 = 3 \\ x_4 = 1 \end{cases}$$

$$\begin{cases} x_3 = 1 - 3x_1 - 4x_2 \\ x_4 = 1 \end{cases} \quad x_1, x_2 = t_1, t_2$$

$$\text{ОП}(t_1, t_2, 1 - 3t_1 - 4t_2, 1) \\ \text{при } t_1 = 0, t_2 = 0 \Rightarrow (0, 0, 1, 1)$$

Дана: сист. коэф. комп. $O.P(L, L, 1-3L+4L, 1)$
 $U.P(0, 0, 1, 1)$

2. 1. 49.

$$\begin{cases} x_1 + x_2 + 3x_3 - 2x_4 + 3x_5 = 1 \\ 2x_1 + 2x_2 + 4x_3 - x_4 + 3x_5 = 2 \\ 3x_1 + 3x_2 + 5x_3 - 2x_4 + 3x_5 = 1 \\ 2x_1 + 2x_2 + 8x_3 - 9x_4 + 9x_5 = 2 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 2 & 2 & 4 & -1 & 3 & 2 \\ 3 & 3 & 5 & -2 & 3 & 1 \\ 2 & 2 & 8 & -9 & 9 & 2 \end{array} \right) \begin{array}{l} \\ \text{II} - 2\text{I} \\ \text{III} - 3\text{I} \\ \text{IV} - 2\text{I} \end{array} \sim$$

$$\sim \left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 & -3 & 0 \\ 0 & 0 & -4 & 4 & -6 & -2 \\ 0 & 0 & 2 & 1 & 3 & 0 \end{array} \right) \begin{array}{l} \\ \\ \text{III} - 2\text{II} \\ \text{IV} + \text{II} \end{array} \sim \left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 & -3 & 0 \\ 0 & 0 & 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 4 & 0 & 0 \end{array} \right) \begin{array}{l} \\ \\ \\ \text{IV} + 2\text{III} \end{array} \sim$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 & -3 & 0 \\ 0 & 0 & 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 & -4 \end{array} \right)$$

$r(A) = 3 \neq r(A|B) = 4 \Rightarrow$ сист. несов. $\Rightarrow$ нет решений

2.2.16

$$f(x) = ax^2 + bx + c$$

$$f(-2) = -8: 4a - 2b + c = -8$$

$$f(1) = 4: a + b + c = 4$$

$$f(2) = -4: 4a + 2b + c = -4$$

$$\begin{cases} 4a - 2b + c = -8 \\ a + b + c = 4 \\ 4a + 2b + c = -4 \end{cases}$$

$$1) A \cdot X = B \Rightarrow X = B \cdot A^{-1}$$

$$A = \begin{vmatrix} 4 & -2 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{vmatrix} = -12 \neq 0 \Rightarrow \exists A^{-1}$$

$$A_{11} = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1$$

$$A_{12} = -\begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = 3$$

$$A_{13} = \begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} = -2$$

$$A_{21} = -\begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} = 4$$

$$A_{22} = \begin{vmatrix} 4 & 1 \\ 4 & 1 \end{vmatrix} = 0$$

$$A_{23} = \begin{vmatrix} 4 & -2 \\ 4 & 2 \end{vmatrix} = -16$$

$$A_{31} = \begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix} = -3$$

$$A_{32} = -\begin{vmatrix} 4 & 1 \\ 1 & 1 \end{vmatrix} = -3$$

$$A_{33} = \begin{vmatrix} 4 & -2 \\ 1 & 1 \end{vmatrix} = 6$$

$$\tilde{A} = \begin{pmatrix} -1 & 4 & -3 \\ 3 & 0 & -3 \\ -2 & -16 & 6 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = -\frac{1}{12} \begin{pmatrix} -1 & 4 & -3 \\ 3 & 0 & -3 \\ -2 & -16 & 6 \end{pmatrix} = \begin{pmatrix} \frac{1}{12} & -\frac{1}{3} & \frac{1}{4} \\ -\frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{6} & \frac{4}{3} & -\frac{1}{2} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{1}{12} & -\frac{1}{3} & \frac{1}{4} \\ -\frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{6} & \frac{4}{3} & -\frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} -8 \\ +4 \\ -4 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ 6 \end{pmatrix}$$

$$D_1 = \begin{vmatrix} -8 & -2 & 1 \\ 4 & 1 & 1 \\ -4 & 2 & 1 \end{vmatrix} = 36$$

$$x_1 = \frac{D_1}{D} = \frac{36}{-12} = -3$$

$$D_2 = \begin{vmatrix} 4 & -8 & 1 \\ 1 & 4 & 1 \\ 4 & -4 & 1 \end{vmatrix} = -12$$

$$x_2 = \frac{D_2}{D} = \frac{-12}{12} = 1$$

$$D_3 = \begin{vmatrix} 4 & -2 & -8 \\ 1 & 1 & 4 \\ 4 & 2 & -4 \end{vmatrix} = -92$$

$$x_3 = \frac{D_3}{D} = \frac{-92}{-12} = 6$$

Ortem: $x_1 = -3$; $x_2 = 1$; $x_3 = 6 \Rightarrow \begin{pmatrix} -3 \\ 1 \\ 6 \end{pmatrix}$

~ 2.2.17.

$$f(x) = a \cdot x^3 + b \cdot x^2 + c$$

$$f(0) = 2: a + c = 2$$

$$f(1) = -1: 3a + b + c = -1$$

$$f(2) = 4: 9a + 4b + c = 4$$

$$\begin{cases} a + c = 2 \\ 3a + b + c = -1 \\ 9a + 4b + c = 4 \end{cases}$$

1) $A \cdot X = B \Rightarrow X = A^{-1} \cdot B$

$$A = \begin{vmatrix} 1 & 0 & 1 \\ 3 & 1 & 1 \\ 9 & 4 & 1 \end{vmatrix} = 0 \Rightarrow \nexists A^{-1}$$

Ответ: нет решения.

2.2.18.

$$\begin{cases} x_1 - x_2 = 5 \\ 2x_1 + x_2 = 1 \end{cases}$$

$$1) AX = B \Rightarrow X = B \cdot A^{-1}$$

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$\det A = 3 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{3} \cdot \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$2) D_1 = \begin{vmatrix} 5 & -1 \\ 1 & 1 \end{vmatrix} = 6$$

$$x_1 = \frac{D_1}{D} = \frac{6}{3} = 2$$

$$D_2 = \begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = -9$$

$$x_2 = \frac{D_2}{D} = \frac{-9}{3} = -3$$

Ответ: $x_1 = 2, x_2 = -3 \Rightarrow \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

~2.2.20.

$$\begin{cases} 2x - y = 2 \\ 2x + dy = 1 \end{cases}$$

$$1) AX = B \Rightarrow X = A^{-1} \cdot B$$

$$\det A = \begin{vmatrix} 2 & -1 \\ 2 & d \end{vmatrix} = 2 \cdot d - (-1) \cdot 2 = 2d + 2 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det A} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{2d+2} \cdot \begin{pmatrix} d & 1 \\ -2 & 2 \end{pmatrix} = \begin{pmatrix} \frac{d}{2d+2} & \frac{1}{2d+2} \\ \frac{-2}{2d+2} & \frac{2}{2d+2} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{d}{2d+2} & \frac{1}{2d+2} \\ \frac{-2}{2d+2} & \frac{2}{2d+2} \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{2d+1}{d^2+2} \\ \frac{d-4}{d^2+2} \end{pmatrix}$$

$$2) D_1 = \begin{vmatrix} 2 & -1 \\ 1 & d \end{vmatrix} = 2d + 1$$

$$x = \frac{D_1}{D} = \frac{2d+1}{d^2+2}$$

$$D_2 = \begin{vmatrix} 2 & 2 \\ 2 & 1 \end{vmatrix} = 2 \cdot 1 - 2 \cdot 2 = 2 - 4$$

$$y = \frac{D_2}{D} = \frac{d-4}{d^2+2}$$

$$\text{Problem: } x = \frac{2d+1}{d^2+2}; y = \frac{d-4}{d^2+2} \Rightarrow \begin{pmatrix} \frac{2d+1}{d^2+2} \\ \frac{d-4}{d^2+2} \end{pmatrix}$$

2.2.24.

$$\begin{cases} 3x + 2y + z = 1 \\ 6x + 5y + 4z = -2 \\ 9x + 8y + 7z = 3 \end{cases}$$

1) $AX = B \Rightarrow X = B \cdot A^{-1}$

$$\det A = \begin{vmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{vmatrix} = 0 \Rightarrow \nexists A^{-1}$$

Ответ: нет решений

2.2.26

$$\begin{cases} ax + by + z = 1 \\ x + a by + z \\ x + by + az = 1 \end{cases}$$

1) $AX = B \Rightarrow X = B \cdot A^{-1}$

$$\det A = \begin{vmatrix} a & b & 1 \\ 1 & ba & 1 \\ 1 & b & a \end{vmatrix} = a \cdot a \cdot b \cdot a + b \cdot 1 \cdot 1 + 1 \cdot 1 \cdot b - 1 \cdot a \cdot b \cdot 1 -$$
$$- 1 \cdot b \cdot a - b \cdot 1 \cdot a = a^3 b - 3ab + 2b =$$
$$= b(a+2) \cdot (a-1)^2 \Rightarrow \text{при}$$

$a = -2$ или $a = 1$ или $b = 0$ - нет решений

при $a \neq -2$ и $a \neq 1$ и $b \neq 0 \Rightarrow$

$$A_{11} = \begin{vmatrix} a & b \\ b & a \end{vmatrix} = a^2 - b^2$$

$$A_{12} = - \begin{vmatrix} 1 & 1 \\ 1 & a \end{vmatrix} = -a + 1$$

$$A_{13} = \begin{vmatrix} 1 & ab \\ 1 & b \end{vmatrix} = b - ab$$

$$A_{21} = - \begin{vmatrix} b & 1 \\ b & a \end{vmatrix} = b - ab$$

$$A_{22} = \begin{vmatrix} a & 1 \\ 1 & a \end{vmatrix} = a^2 - 1$$

$$A_{23} = - \begin{vmatrix} a & b \\ 1 & b \end{vmatrix} = b - ab$$

$$A_{31} = \begin{vmatrix} b & 1 \\ ab & 1 \end{vmatrix} = b - ab$$

$$A_{32} = - \begin{vmatrix} a & 1 \\ 1 & 1 \end{vmatrix} = 1 - a$$

$$A_{33} = \begin{vmatrix} a & b \\ 1 & ab \end{vmatrix} = a^2 b - b$$

$$\tilde{A} = \begin{pmatrix} a^2 b - b & b - ab & b - ab \\ 1 - a & a^2 - 1 & 1 - a \\ b - ab & b - ab & a^2 b - b \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{a^3 b - 5ab + 1} \cdot \begin{pmatrix} a^2 b - b & b - ab & b - ab \\ 1 - a & a^2 - 1 & 1 - a \\ b - ab & b - ab & a^2 b - b \end{pmatrix}$$

$$= \begin{pmatrix} \frac{a^1 b - b}{a^3 b - 3ab + 1b} & \frac{b - ab}{a^3 b - 3ab + 1b} & \frac{a^3 b - 3ab + 1b}{a^3 b - 3ab + 1b} \\ \frac{1-a}{a^3 b - 3ab + 1b} & \frac{a^1 - 1}{a^3 b - 3ab + 1b} & \frac{1-a}{a^3 b - 3ab + 1b} \\ \frac{b - ab}{a^3 b - 3ab + 1b} & \frac{b - ab}{a^3 b - 3ab + 1b} & \frac{a^1 b - b}{a^3 b - 3ab + 1b} \end{pmatrix} =$$

$$\begin{pmatrix} \frac{a+1}{a^1 + a - 2} & \frac{-1}{a^1 + a - 2} & \frac{-1}{a^1 + a - 2} \\ \frac{-1}{a^1 b + ab - 2b} & \frac{a+1}{a^1 b + ab - 2b} & \frac{-1}{a^1 b + ab - 2b} \\ \frac{-1}{a^1 + a - 2} & \frac{1}{a^1 + a - 2} & \frac{a+1}{a^1 + a - 2} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{a-b}{a^2 + a - 1} \\ \frac{ab + b - 2}{a^1 b + ab - 2b} \\ \frac{a-b}{a^1 + a - 1} \end{pmatrix}$$

2) $D = \det A = a^3 b - 3ab + 2b$

$$D_1 = \begin{vmatrix} 1 & b & 1 \\ b & ab & 1 \\ 1 & b & a \end{vmatrix} = 1 \cdot ab \cdot a + b \cdot 1 \cdot 1 + b \cdot b \cdot 1 - 1 \cdot ab \cdot 1 - b \cdot b \cdot a - b \cdot 1 \cdot 1 = a^2 b + b^2 - ab - ab^2$$

$$x = \frac{D_1}{D} = \frac{a^2 b + b^2 - ab - ab^2}{a^3 b - 3ab + 2b} = \frac{b(a+1)(a-b)}{b(a+2)(a-1)^2} = \frac{a-b}{(a+2)(a-1)^2}$$

$$D_2 = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & a \end{vmatrix} = a \cdot b \cdot a + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 1 - 1 \cdot b \cdot 1 - 1 \cdot 1 \cdot a - 1 \cdot 1 \cdot a = a^2b - 2a - b + 2$$

$$y = \frac{D_2}{D} = \frac{a^2b - 2a - b + 2}{a^3b - 3ab + 2b} = \frac{ab + b - 1}{a^2b - ab - 2b}$$

$$D_3 = \begin{vmatrix} a & b & 1 \\ 1 & a & b \\ 1 & b & 1 \end{vmatrix} = a \cdot a \cdot b + b \cdot b \cdot 1 + 1 \cdot 1 \cdot b - 1 \cdot a \cdot b - 1 \cdot b \cdot 1 - 1 \cdot b \cdot a = a^2b + b^2 - ab - ab = a^2b + b^2 - 2ab$$

$$z = \frac{D_3}{D} = \frac{a^2b + b^2 - 2ab}{a^3b - 3ab + 2b} = \frac{a - b}{a^2 + a - 1}$$

Orbital: $x = \frac{a-b}{a^2+a-1}$
 $y = \frac{ab+b-1}{a^2+ab-2b}$
 $z = \frac{a-b}{a^2+a-1}$

$$\Rightarrow \begin{pmatrix} \frac{a-b}{a^2+a-1} \\ \frac{ab+b-1}{a^2+ab-2b} \\ \frac{a-b}{a^2+a-1} \end{pmatrix}$$

~2.2.31

$$f(x) = a \cdot \log_3 x + bx + c$$

$$f(1) = 5: a \cdot \log_3 1 + b \cdot 1 + c = 5$$

$$a \cdot 0 + b + c = 5$$

$$b + c = 5$$

$$f(3) = 8: a \cdot \log_3 3 + b \cdot 3 + c = 8$$

$$a \cdot 1 + 3b + c = 8$$

$$a + 3b + c = 8$$

$$f(9) = 19: a \cdot 9 + b \cdot 9 + c = 19$$

$$a \cdot 2 + 9b + c = 19$$

$$2a + 9b + c = 19$$

$$\begin{cases} b + c = 5 \\ a + 3b + c = 8 \\ 2a + 9b + c = 19 \end{cases}$$

$$1) AX = B \Rightarrow X = B \cdot A^{-1}$$

$$\det A = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 3 & 1 \\ 2 & 9 & 1 \end{vmatrix} = 4 \Rightarrow \exists A^{-1}$$

$$A_{11} = \begin{vmatrix} 3 & 1 \\ 9 & 1 \end{vmatrix} = -8$$

$$A_{12} = - \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1$$

$$A_{13} = \begin{vmatrix} 1 & 3 \\ 2 & 9 \end{vmatrix} = 3$$

$$A_{21} = - \begin{vmatrix} 1 & 1 \\ 9 & 1 \end{vmatrix} = 8$$

$$A_{22} = \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = -2$$

$$A_{23} = - \begin{vmatrix} 0 & 1 \\ 2 & 9 \end{vmatrix} = 2$$

$$A_{31} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = -2$$

$$A_{32} = - \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1$$

$$A_{33} = \begin{vmatrix} 0 & 1 \\ 1 & 3 \end{vmatrix} = -1$$

$$\tilde{A} = \begin{pmatrix} -6 & 8 & -2 \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{4} \cdot \begin{pmatrix} -6 & 8 & -2 \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} & 2 & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{2} & -\frac{1}{4} \end{pmatrix}$$

$$X = A^{-1} \cdot B = \begin{pmatrix} -\frac{3}{2} & 2 & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{2} & -\frac{1}{4} \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 8 \\ 19 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$$

$$2) D_1 = \begin{vmatrix} 5 & 1 & 1 \\ 8 & 3 & 1 \\ 19 & 9 & 1 \end{vmatrix} = -4$$

$$\alpha_1 = \frac{D_1}{D} = \frac{-4}{4} = -1$$

$$D_2 = \begin{vmatrix} 0 & 5 & 1 \\ 1 & 8 & 1 \\ 2 & 19 & 1 \end{vmatrix} = 8$$

$$\alpha_2 = \frac{D_2}{D} = \frac{8}{4} = 2$$

$$D_3 = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 3 & 8 \\ 2 & 9 & 19 \end{vmatrix} = 12$$

$$x_3 = \frac{D_3}{D} = \frac{12}{4} = 3$$

Ответ: $x_1 = 1$
 $x_2 = 2$
 $x_3 = 3 \Rightarrow \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$

2.3.26

$$\begin{cases} 2x_1 - x_2 = 0 \\ -4x_1 + 2x_2 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 2 & -1 & 0 \\ -4 & 2 & 0 \end{array} \right) \xrightarrow{\Pi + 2I} \left(\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{C.C.} \mid \rightarrow \text{C неопр.}$$

$$n = 2 > r$$

$$|a_{11}| = |2| = 2 \neq 0 \Rightarrow x_1 - \text{н. перем.}$$

$$x_2 - \text{св. перем.}$$

$$2x_1 - x_2 = 0$$

$$2x_1 = x_2$$

$$x_1 = \frac{1}{2} x_2$$

$$\exists x_2 = t \Rightarrow x_1 = \frac{t}{2}$$

$$\text{О.Р. } \left(\frac{t}{2}; t \right)$$

Ф.С.Р. описан CЛAУ

$$\left(\frac{t}{2}; t \right) = \left(\frac{1}{2} \cdot t; 1 \cdot t \right) = t \left(\frac{1}{2}; 1 \right) = \left\{ \left(\frac{1}{2}; 1 \right) \right\}$$

Омлем: О.Р. $(\frac{1}{\sqrt{3}}; t)$

Ф.Л.Р. опор. (Л.А.У: $\{(\frac{1}{\sqrt{3}}; 1)\}$

~ 23.24

$$\begin{cases} x - \sqrt{3}y = 0 \\ \sqrt{3}x - 3y \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 1 & -\sqrt{3} & 0 \\ \sqrt{3} & -3 & 0 \end{array} \right) \xrightarrow{\text{II} - \sqrt{3}\text{I}} \left(\begin{array}{cc|c} 1 & -\sqrt{3} & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{Л.Л.}$$

$$n = 2 > r \Rightarrow \text{Л. неопр.}$$

$$r(A) = 1 \Rightarrow 1 \text{ л. перем.}$$

$$n - r = 2 - 1 = 1 \Rightarrow 1 \text{ свобод. перем.}$$

$$|a_{11}| = |1| = 1 \neq 0 \Rightarrow$$

$$\Rightarrow x_1 - \text{л. перем.}$$

$$x_2 - \text{св. перем.}$$

$$x_1 - \sqrt{3}x_2 = 0$$

$$x_1 = \sqrt{3}x_2$$

$$\exists x_2 = \frac{1}{\sqrt{3}} t \Rightarrow x_1 = \sqrt{3} \cdot \frac{1}{\sqrt{3}} \cdot t = t$$

$$\text{О.Р. } (t; \frac{1}{\sqrt{3}} t)$$

$$\text{Ф.Л.Р. } (t; \frac{1}{\sqrt{3}} t) = (1 \cdot t; \frac{1}{\sqrt{3}} t) = t \left(1; \frac{1}{\sqrt{3}} \right) = \left\{ \left(1; \frac{1}{\sqrt{3}} \right) \right\}$$

Омлем: О.Р. $(\frac{1}{\sqrt{3}}; t)$

Ф.Л.Р. опор. (Л.А.У: $\{(\frac{1}{\sqrt{3}}; 1)\}$)

~ 23.24

$$\begin{cases} x - \sqrt{3}y = 0 \\ \sqrt{3}x - 3y \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 1 & -\sqrt{3} & 0 \\ \sqrt{3} & -3 & 0 \end{array} \right) \xrightarrow{\text{II} - \sqrt{3}\text{I}} \left(\begin{array}{cc|c} 1 & -\sqrt{3} & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{Л.Л.}$$

$$n = 2 > r \Rightarrow \text{Л. неопр.}$$

$$r(A) = 1 \Rightarrow 1 \text{ л. перем.}$$

$$n - r = 2 - 1 = 1 \Rightarrow 1 \text{ свобод. перем.}$$

$$|a_{11}| = |1| = 1 \neq 0 \Rightarrow$$

$$\Rightarrow x_1 - \text{л. перем.}$$

$$x_2 - \text{св. перем.}$$

$$x_1 - \sqrt{3}x_2 = 0$$

$$x_1 = \sqrt{3}x_2$$

$$\exists x_2 = \frac{1}{\sqrt{3}} t \Rightarrow x_1 = \sqrt{3} \cdot \frac{1}{\sqrt{3}} \cdot t = t$$

$$\text{О.Р. } (t; \frac{1}{\sqrt{3}} t)$$

$$\text{Ф.Л.Р. } (t; \frac{1}{\sqrt{3}} t) = (1 \cdot t; \frac{1}{\sqrt{3}} t) = t \left(1; \frac{1}{\sqrt{3}} \right) = \left\{ \left(1; \frac{1}{\sqrt{3}} \right) \right\}$$

$$\text{Omlen: O.P.: } (t; \frac{1}{\sqrt{3}}t)$$

$$\text{O.C.P.: } \{(1; \frac{1}{\sqrt{3}})\}$$

$$\sim 2.3.28.$$

$$\begin{cases} 3x + 4y = 0 \\ 4x - 3y = 0 \end{cases}$$

$$\begin{cases} 4x - 3y = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 3 & 4 & 0 \\ 4 & -3 & 0 \end{array} \right) \xrightarrow{3II - 4I} \left(\begin{array}{cc|c} 3 & 4 & 0 \\ 0 & -25 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 2 \Rightarrow \text{C.C.}$$

$$n=2 \Rightarrow r \Rightarrow \text{C. onp.}$$

$$\begin{cases} 3x + 4y = 0 \\ -25y = 0 \end{cases} \quad \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$$\text{O.P.: } (0; 0)$$

$$\text{O.C.P.} = \emptyset$$

$$\text{Omlen: O.P. } (0; 0), \text{ O.C.P. } = \emptyset$$

$$\sim 2.3.29$$

$$\begin{cases} x_1 + 2x_2 = 0 \\ -\sqrt{3}x_1 - \sqrt{12}x_2 = 0 \\ 2x_1 + 4x_2 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cc|c} 1 & 2 & 0 \\ -\sqrt{3} & -\sqrt{12} & 0 \\ 2 & 4 & 0 \end{array} \right) \xrightarrow{\begin{array}{l} II + \sqrt{3}I \\ III - 2I \end{array}} \sim$$

$$\sim \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{L. C.}$$

$$n = 2 > r \Rightarrow \text{L. неопр.}$$

$$r(A) = 1 \Rightarrow 1 \text{ л. независ.}$$

$$n - r = 2 - 1 = 1 \Rightarrow 1 \text{ св. завис.}$$

$$|a_{11}| = 1 \Rightarrow x_1 - \text{л. независ.}$$

$$\text{сл. св. завис.}$$

$$x_1 + 2x_2 = 0$$

$$\text{сл. } 1 = -2\text{сл. } 2$$

$$\] x_2 = t \Rightarrow x_1 = -2t$$

$$\text{О. П. } (-2t; t)$$

$$\text{П. С. П. } (-2t; t) = (-2 \cdot t; 1 \cdot t) = t(-2; 1) = \{(-2; 1)\}$$

$$\text{Ответ: О. П. } (-2t; t)$$

$$\text{П. С. П. } \{(-2; 1)\}$$

$$r 2. 3. 30$$

$$\begin{cases} 2x - y - z = 0 \\ 4x - 2y - 2z = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 4 & -2 & -1 & 0 \end{array} \right) \xrightarrow{II-2I} \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{C.C.}$$

$$n = 3 > r$$

$$r(A) = 1 \Rightarrow 1 \text{ w. repen.}$$

$$n - r = 3 - 1 = 2 \Rightarrow 2 \text{ f. repen.}$$

$$|a_{ii}| = |2| = 2 \Rightarrow x \text{ - w. repen.}$$

$$y, z \text{ - f. repen.}$$

$$2x = y + z$$

$$x = \frac{1}{2}(y + z) = \frac{1}{2}y + \frac{1}{2}z$$

$$\exists y = 2t_1 \Rightarrow x = t_1 + t_2$$

$$z = 2t_2$$

$$\text{O.P.} : (t_1 + t_2; 2t_1; 2t_2)$$

$$\text{G.C.P.} : (t_1 + t_2; 2t_1; 2t_2) = (1 \cdot t_1 + 1 \cdot t_2; 2 \cdot t_1; 2 \cdot t_2)$$

$$= (1 \cdot t_1; 2 \cdot t_1; 0 \cdot t_1) + (1 \cdot t_2; 0 \cdot t_2; 2 \cdot t_2)$$

$$= t_1(1; 2; 0) + t_2(1; 0; 2) = \{(1; 2; 0); (1; 0; 2)\}$$

$$\text{Ambem: O.P.} : (t_1 + t_2; 2t_1; 2t_2)$$

$$\text{G.C.P.} : \{(1; 2; 0); (1; 0; 2)\}$$

$$\sim 2.3.31.$$

$$\begin{cases} 2x - y - z = 0 \\ x + 2y + 3z = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 1 & 2 & 3 & 0 \end{array} \right) \xrightarrow{2 \cdot II - I} \left(\begin{array}{ccc|c} 2 & -1 & -1 & 0 \\ 0 & 5 & 4 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 2 \Rightarrow \text{L.C.}$$

$$n = 3 \Rightarrow r = \text{L. neopn.}$$

$$r(A) = 2 \Rightarrow 2 \text{ ne. neopn.}$$

$$n - r = 3 - 2 = 1 - 1 \text{ sv. neopn.}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 0 & 5 \end{vmatrix} = 2 \cdot 5 - (-1) \cdot 0 = 10 \neq 0$$

x, y - sv. neopn.

z - sv. neopn.

$$\begin{cases} 2x - y - z = 0 \\ 5y + 4z = 0 \end{cases} \quad \begin{cases} 2x - y - z = 0 \\ 5y = -4z \end{cases}$$

$$\begin{cases} 2x - y - z = 0 \\ y = -\frac{4}{5}z \end{cases} \quad \begin{cases} 2x + \frac{4}{5}z - z = 0 \\ y = -\frac{4}{5}z \end{cases}$$

$$\begin{cases} 2x = -\frac{2}{5}z \\ y = -\frac{4}{5}z \end{cases} \quad \begin{cases} x = -\frac{1}{5}z \\ y = -\frac{4}{5}z \end{cases}$$

$$\Rightarrow z = -5t, \text{ maka}$$

$$\begin{cases} x = t \\ y = 4t \end{cases}$$

Общая решение: $(t; 4t; -5t)$
 $\mathcal{O.L.P.}: (t; 4t; -5t) = (1 \cdot t; 4 \cdot t; -5 \cdot t) =$
 $= t(1; 4; -5) = \{ (1; 4; -5) \}$
 Ответ: О.Р.: $(1; 4; -5)$
 $\mathcal{O.L.P.}: \{ (1; 4; -5) \}$

~ 2.3.32

$$\begin{cases} 3x_1 - 2x_2 + x_3 = 0 \\ 2x_1 + 5x_2 + 3x_3 = 0 \\ 3x_1 + 4x_2 + 2x_3 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 3 & -2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ 3 & 4 & 2 & 0 \end{array} \right) \begin{array}{l} 3 \cdot II - 2 \cdot I \sim \\ III - I \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 3 & -2 & 1 & 0 \\ 0 & 19 & 7 & 0 \\ 0 & 6 & 1 & 0 \end{array} \right) \xrightarrow{19 III - 6 II} \sim \left(\begin{array}{ccc|c} 3 & -2 & 1 & 0 \\ 0 & 19 & 7 & 0 \\ 0 & 0 & -23 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 3 \Rightarrow \text{C.C.}$$

$$n = 3 \geq r \Rightarrow \text{C. Опр.}$$

$$\begin{cases} 3x_1 - 2x_2 + x_3 = 0 \\ 19x_2 + 7x_3 = 0 \\ -23x_3 = 0 \end{cases} \begin{cases} 3x_1 - 2x_2 = 0 \\ 19x_2 = 0 \\ x_3 = 0 \end{cases} \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

$$\text{од. } \mathcal{P}: (0; 0; 0)$$

$$\mathcal{P} \cap \mathcal{L} \cap \mathcal{P} = \emptyset$$

$$\text{антем: од. } \mathcal{P}: (0; 0; 0)$$

$$\mathcal{P} \cap \mathcal{L} \cap \mathcal{P} = \emptyset$$

$$\sim 2.3.33.$$

$$\begin{cases} x_1 - 2x_2 - 3x_3 = 0 \\ 2x_1 + 3x_2 + x_3 = 0 \\ 5x_1 - 3x_2 - 8x_3 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 2 & 3 & 1 & 0 \\ 5 & -3 & -8 & 0 \end{array} \right) \begin{array}{l} \\ \text{II} - 2\text{I} \\ \text{III} - 5\text{I} \end{array} \sim \left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & 7 & 7 & 0 \\ 0 & 7 & 7 & 0 \end{array} \right) \begin{array}{l} \\ \\ \text{III} - \text{II} \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & 7 & 7 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 2 - \text{Л.С.}$$

$$n = 3 > r - \text{сист. неопр}$$

$$r(A) = 2 \Rightarrow 2 \text{ зл. перемен.}$$

$$n - r = 3 - 2 = 1 \Rightarrow 1 \text{ св. перемен}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 1 & -2 \\ 0 & 4 \end{vmatrix} = 1 \cdot 4 - (-2) \cdot 0 = 4 \neq 0 \Rightarrow$$

$\Rightarrow x_1, x_2$ — независ.
 x_3 — завис.

$$\begin{cases} x_1 - 2x_2 - 3x_3 = 0 \\ 4x_2 + 4x_3 = 0 \end{cases}$$

$$\begin{cases} x_1 - 2x_2 - 3x_3 = 0 \\ 4x_2 = -4x_3 \end{cases}$$

$$\begin{cases} x_1 + 2x_3 - 3x_3 = 0 \\ x_2 = -x_3 \end{cases}$$

$$\begin{cases} x_1 = x_3 \\ x_2 = -x_3 \end{cases}$$

$$\exists x_3 = t$$

Пусть, $x_1 = t$
 $x_2 = -t$

О.П. $(t; -t; t)$

Ф.С.П.: $(t; -t; t) = (1 \cdot t; (-1) \cdot t; 1 \cdot t) = t(1; -1; 1) =$
 $= \{ (1; -1; 1) \}$

Ответ: О.П. $(t; -t; t)$

Ф.С.П.: $\{ (1; -1; 1) \}$

~ 2.3.39.

$$\begin{cases} x_1 - 2x_2 + 3x_3 = 0 \\ -x_1 + 2x_2 - 3x_3 = 0 \\ 2x_1 - 4x_2 + 6x_3 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ -1 & 2 & -3 & 0 \\ 2 & -4 & 6 & 0 \end{array} \right) \begin{array}{l} \\ \text{II} + \text{I} \\ \text{III} - 2 \cdot \text{I} \end{array} \sim \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 1 \Rightarrow \text{умм. лобн}$$

$$n = 3 > r \Rightarrow \text{умм. неопр.}$$

$$r(A) = 1 \Rightarrow 1 \text{ л. неперн}$$

$$n - r = 3 - 1 = 2 \Rightarrow 2 \text{ лб. неперн.}$$

$$|a_{11}| = |1| = 1 \neq 0$$

$$x_1 - \text{л. неперн.}$$

$$x_2, x_3 - \text{лб. неперн.}$$

$$x_1 - 2x_2 + 3x_3 = 0$$

$$x_1 = 2x_2 - 3x_3$$

$$\left. \begin{array}{l} x_2 = t_1 \\ x_3 = t_2 \end{array} \right\} \Rightarrow x_1 = 2t_1 - 3t_2$$

$$x_3 = t_2$$

$$\text{О.Р. } (2t_1 - 3t_2; t_1; t_2)$$

$$\text{В.С.Р. } (2t_1 - 3t_2; t_1; t_2) = (1 \cdot 2 \cdot t_1 + (-3) \cdot t_2;$$

$$1 \cdot t_1 + 0 \cdot t_2; 0 \cdot t_1 + 1 \cdot t_2) = (2 \cdot t_1; 1 \cdot t_1; 0 \cdot t_1) +$$

$$+ t_2(-3; 0; 1) = \{(2; 1; 0), (-3; 0; 1)\}$$

$$\text{GIVEN: O.P.: } (2t_1 - 3t_2; t_1; t_1)$$

$$\text{P.L.O. system: } \{(2; 1; 0), (1; 3; 0; 1)\}$$

$$\sim 23.35$$

$$\begin{cases} 2x_1 + x_2 - x_3 = 0 \\ x_1 - 2x_2 + x_3 = 0 \\ x_1 + 3x_2 - 2x_3 = 0 \\ x_1 + 8x_2 - 5x_3 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 3 & -2 & 0 \\ 1 & 8 & 5 & 0 \end{array} \right) \begin{array}{l} 2\text{II} - \text{I} \\ 2\text{III} - \text{I} \\ 2\text{IV} - \text{I} \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 15 & 11 & 0 \end{array} \right) \begin{array}{l} \text{III} + \text{II} \\ \text{IV} + 3\text{II} \end{array} \sim \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 20 & 0 \end{array} \right) \begin{array}{l} \text{III} \leftrightarrow \text{IV} \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 0 & 20 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 3 \Rightarrow \text{L.C.}$$

$$n=3=r \Rightarrow \text{unim. sys.}$$

$$\begin{cases} 2x_1 + x_2 - x_3 = 0 \\ -5x_2 + 3x_3 = 0 \\ 20x_3 = 0 \end{cases} \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

Общее решение: $(0; 0; 0)$

О.Л.Р. - $\emptyset$

~ 2.3.36.

$$\begin{cases} x_1 - x_3 + x_5 = 0 \\ x_2 - x_4 + x_6 = 0 \\ x_1 - x_2 + x_5 - x_6 = 0 \\ x_2 - x_3 + x_6 = 0 \\ x_1 - x_4 + x_5 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccccc|c} 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 & 1 & 0 & 0 \end{array} \right) \begin{array}{l} \\ \\ \text{III} - \text{I} + \text{II} \\ \text{IV} - \text{I} + \text{II} \\ \end{array} \sim$$

$$\sim \left(\begin{array}{cccccc|c} 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 \end{array} \right) \sim \begin{array}{l} \\ \\ \\ \text{IV} + \text{III} \\ \text{V} - \text{III} \end{array}$$

$$\sim \left(\begin{array}{cccccc|c} 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 3 \Rightarrow \text{L.C.}$$

$$n=6 > r \Rightarrow \text{L. kerne}$$

$$r(A)=3 \Rightarrow 3 \text{ w. nepen.}$$

$$n-r=6-3=3 \Rightarrow 3 \text{ cb. nepen.}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot 1 \cdot 1 + 0 \cdot 0 \cdot 0 + 0 \cdot 0 \cdot (-1) -$$

$$- 0 \cdot 1 \cdot (-1) - 0 \cdot 0 \cdot 1 - 0 \cdot 0 \cdot 1 =$$

$$1 \neq 0 \Rightarrow$$

$$\Rightarrow x_1, x_2, x_3 - \text{w. nepen.}$$

$$x_4, x_5, x_6 - \text{cb. nepen.}$$

$$\begin{cases} x_1 - x_3 + x_5 = 0 \\ x_2 - x_4 + x_6 = 0 \\ x_3 - x_4 = 0 \end{cases} \quad \begin{cases} x_1 - x_4 + x_5 = 0 \\ x_2 = x_4 - x_6 \\ x_3 = x_4 \end{cases}$$

$$\begin{cases} x_1 = x_4 - x_5 \\ x_2 = x_4 - x_6 \\ x_3 = x_4 \end{cases}$$

$$\left. \begin{array}{l} x_4 = t_1 \\ x_5 = t_2 \\ x_6 = t_3 \end{array} \right\} \Rightarrow \begin{cases} x_1 = t_1 - t_2 \\ x_2 = t_1 - t_3 \\ x_3 = t_1 \end{cases}$$

$$O.P.: (t_1 - t_2; t_1 - t_3; t_1; t_1; t_2; t_3)$$

$$\begin{aligned} \mathcal{P.C.P.:} (t_1 - t_2; t_1 - t_3; t_1; t_1; t_2; t_3) &= \\ &= (1 \cdot t_1 - 1 \cdot t_2 + 0 \cdot t_3; 1 \cdot t_1 - 1 \cdot t_3 + 0 \cdot t_2; 1 \cdot t_1 + \\ &+ 0 \cdot t_2 + 0 \cdot t_3; 1 \cdot t_1 + 0 \cdot t_2 + 0 \cdot t_3; 0 \cdot t_1 + 1 \cdot t_2 + \\ &0 \cdot t_3; 0 \cdot t_1 + 0 \cdot t_2 + 1 \cdot t_3) = (1 \cdot t_1 + (-1) \cdot t_2 + \\ &+ 0 \cdot t_3; 1 \cdot t_1 + 0 \cdot t_2 + (-1) \cdot t_3; 1 \cdot t_1 + 0 \cdot t_2 + 0 \cdot t_3; \\ &1 \cdot t_1 + 0 \cdot t_2 + 0 \cdot t_3; 0 \cdot t_1 + 1 \cdot t_2 + 0 \cdot t_3; 0 \cdot t_1 + 0 \cdot t_2 \\ &+ 1 \cdot t_3) = (1 \cdot t_1; 1 \cdot t_1; 1 \cdot t_1; 1 \cdot t_1; 0 \cdot t_1; 0 \cdot t_1) + \\ &+ ((-1) \cdot t_2; 0 \cdot t_2; 0 \cdot t_2; 0 \cdot t_2; 1 \cdot t_2; 0 \cdot t_2) + \\ &+ (0 \cdot t_3; (-1) \cdot t_3; 0 \cdot t_3; 0 \cdot t_3; 0 \cdot t_3; 1 \cdot t_3) = \end{aligned}$$

$$= t_1 \cdot (1, 1, 1, 1, 0, 0) + t_2 \cdot (-1, 0, 0, 0, 1, 0) + t_3 \cdot (0, -1, 0, 0, 0, 1) = \{(1, 1, 1, 1, 0, 0), (-1, 0, 0, 0, 1, 0), (0, -1, 0, 0, 0, 1)\}$$

Orbem: O. P.: $(t_1 - t_2, t_1 - t_3, t_1, t_2, t_3)$

O. L. P.: $\{(1, 1, 1, 1, 0, 0), (-1, 0, 0, 0, 1, 0), (0, -1, 0, 0, 0, 1)\}$

~ 23.57

$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 0 \\ x_1 + 3x_2 - 3x_3 + 4x_4 = 0 \\ 3x_1 + 2x_2 + x_3 = 0 \\ x_1 + 3x_2 - 5x_4 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 1 & 3 & -3 & 4 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & -5 & 0 \end{array} \right) \begin{array}{l} \\ II - I \\ III - 3I \\ IV - I \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & -1 & 4 & -6 & 0 \\ 0 & 2 & 1 & -7 & 0 \end{array} \right) \begin{array}{l} \\ \\ 2 \cdot III + II \\ IV - II \end{array} \sim \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & 0 & 6 & -10 & 0 \\ 0 & 0 & 3 & -9 & 0 \end{array} \right) \begin{array}{l} \\ \\ \\ 2 \cdot IV - III \end{array} \sim$$

$$\sim \begin{pmatrix} 1 & 1 & -1 & 2 & | & 0 \\ 0 & 2 & -2 & 2 & | & 0 \\ 0 & 0 & 6 & -10 & | & 0 \\ 0 & 0 & 0 & -8 & | & 0 \end{pmatrix}$$

$$r(A) = r(A|B) = 4 \Rightarrow \text{L.C.}$$

$$n = 4 = r \Rightarrow \text{L. o. n.}$$

$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 0 \\ 2x_2 - 2x_3 + 2x_4 = 0 \\ 6x_3 - 10x_4 = 0 \\ -8x_4 = 0 \end{cases} \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

$$\text{O.P. } (0; 0; 0; 0)$$

$$\text{O.C.P.} = \emptyset$$

$$\text{Ombem: O.P. } (0; 0; 0; 0)$$

$$\text{O.C.P.} = \emptyset$$

$$2.3.38.$$

$$\begin{cases} 2x_1 - 4x_2 + 5x_3 + 3x_4 = 0 \\ 3x_1 - 6x_2 + 4x_3 + 1x_4 = 0 \\ 4x_1 - 8x_2 + 17x_3 + 11x_4 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccc|c} 2 & -4 & 5 & 3 & 0 \\ 3 & -6 & 4 & 2 & 0 \\ 4 & -8 & 17 & 11 & 0 \end{array} \right) \begin{array}{l} 2 \cdot II - 3I \\ III - 2I \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|c} 2 & -4 & 5 & 3 & 0 \\ 0 & 0 & -7 & -5 & 0 \\ 0 & 0 & 7 & 5 & 0 \end{array} \right) \begin{array}{l} \\ \\ III + II \end{array} \sim \left(\begin{array}{cccc|c} 2 & -4 & 5 & 3 & 0 \\ 0 & 0 & -7 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$r(A) = r(A|B) = 2 \neq C.C.$$

$$n=4 > r \Rightarrow C. \text{ не пер.}$$

$$r(A)=2 \Rightarrow 2 \text{ л. пер.}$$

$$n-r = 4-2 = 2 - 2 \text{ св. перем.}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 2 & -4 \\ 0 & 0 \end{vmatrix} = 0$$

$$\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} = \begin{vmatrix} -4 & 5 \\ 0 & -7 \end{vmatrix} = 28 \Rightarrow$$

$$\Rightarrow x_2, x_3 - \text{ л. пер.}$$

$$x_1, x_4 - \text{ св. перем.}$$

$$\begin{cases} 2x_1 - 4x_2 + 5x_3 + 3x_4 = 0 \\ -7x_3 - 5x_4 \end{cases}$$

$$\begin{cases} 2x_1 - 4x_2 + 5x_3 = 0 \\ x_3 = -\frac{5}{7}x_4 \end{cases}$$

$$2x_1 - 4x_2 - \frac{4}{7}x_4 = 0$$

$$-4x_2 = \frac{4}{7}x_4 - 2x_1$$

$$4x_2 = 2x_1 - \frac{4}{7}x_4$$

$$x_2 = \frac{1}{2}x_1 - \frac{1}{7}x_4 \Rightarrow \begin{cases} x_2 = \frac{1}{2}x_1 - \frac{1}{7}x_4 \\ x_3 = -\frac{5}{7}x_4 \end{cases}$$

$$\left] \begin{cases} x_1 = 2t_1 \\ x_4 = 7t_2 \end{cases} \Rightarrow \begin{cases} x_2 = t_1 - t_2 \\ x_3 = -5t_2 \end{cases}$$

$$O.P. (2t_1; t_1 - t_2; -5t_2; 7t_2)$$

$$O.C. P. (2t_1; t_1 - t_2; -5t_2; 7t_2) = (2t_1 + 0 \cdot t_2;$$

$$1 \cdot t_1 + (-1) \cdot t_2; 0 \cdot t_1 + (-5) \cdot t_2; 0 \cdot t_1 + 7 \cdot t_2) = (2t_1; t_1;$$

$$0 \cdot t_1; 0 \cdot t_1) + (0 \cdot t_2; (-1) \cdot t_2; (-5) \cdot t_2; 7 \cdot t_2) =$$

$$= t_1(2; 1; 0; 0) + t_2(0; -1; -5; 7) = \{(2; 1; 0; 0),$$

$$(0; -1; -5; 7)\}$$

$$\text{Ornbern: } O.P. (2t_1; t_1 - t_2; -5t_2; 7t_2)$$

$$O.C.P. \{ (2; 1; 0; 0), (0; -1; -5; 7) \}$$

2.3.40

$$\begin{cases} 2x_1 + x_2 + 3x_3 = 0 \\ 4x_1 - x_2 + 7x_3 = 0 \\ x_1 + 2x_2 + 2x_3 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{ccc|c} 2 & 1 & 3 & 0 \\ 4 & -1 & 7 & 0 \\ -1 & a & 2 & 0 \end{array} \right) \begin{array}{l} \text{II} - 2\text{I} \\ 2 \cdot \text{III} - \text{I} \end{array} \sim \left(\begin{array}{ccc|c} 2 & 1 & 3 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 2a+1 & -4 & 0 \end{array} \right) \begin{array}{l} \\ \\ 3 \cdot \text{II} + 12a - 4 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 2 & 1 & 3 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 2+4a & 0 \end{array} \right)$$

1) Пусть $a + 2a = 0$

$$1 + a = 0$$

$$a = -1$$

$$r(A) = r(A|B) = 2 = \text{числ. ранг.}$$

$$n = 3 > r \Rightarrow \text{система некорр.}$$

$$r(A) = 2 \Rightarrow 2 \text{ л. перем.}$$

$$n - r = 3 - 2 = 1 - \text{своб. перем.}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 0 & -3 \end{vmatrix} = 2(-3) - 1 \cdot 0 = -6 \Rightarrow$$

$$\Rightarrow x_1, x_2 - \text{зав. перм.}$$

$$x_3 - \text{св. перм.}$$

$$\begin{cases} 2x_1 + x_2 + 3x_3 = 0 \\ -3x_2 + x_3 = 0 \end{cases}$$

$$\begin{cases} 2x_1 + x_2 + 3x_3 = 0 \\ x_2 = -\frac{1}{3}x_3 \end{cases}$$

$$\begin{cases} x_1 = -\frac{5}{3}x_3 \\ x_2 = \frac{1}{3}x_3 \end{cases}$$

$$\text{]} } x_3 = 3t \Rightarrow \begin{cases} x_1 = -5t \\ x_2 = t \end{cases}$$

$$O.P. (-5t; t; 3t)$$

$$O.C.P. (-5t; t; 3t) = (-5 \cdot t; t; 3 \cdot t) = t(-5; 1; 3) =$$

$$= \{(-5; 1; 3)\}$$

$$\text{Типу } 2 + 2a \neq 0$$

$$1 + a \neq 0$$

$$a \neq -1$$

$$r(A) = r(A|B) = 3 - C.C.$$

$$n = r = 3 \Rightarrow \text{C. onp.}$$

$$\begin{cases} 2x_1 + x_2 + 3x_3 = 0 \\ -3x_1 + 2x_3 = 0 \\ (1+a)x_3 = 0 \end{cases}$$

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

$$O.P. (0; 0; 0)$$

$$O.C.P. \emptyset$$

Ответа: Типу $a = -1$ - система совместна,

homog. O.P. $(-5t; t; 3t)$, P.C.P. $\{(-5, 1, 3)\}$
 type $a \neq 1$ - unique solution, orig. O.P.
 $(0, 0, 0)$, P.C.P. $= \emptyset$
 $\sim 2.3.41.$

$$\begin{cases}
 x_1 - 3x_2 + x_3 - 2x_4 = 0 \\
 3x_1 + 2x_2 + 3x_4 = 0 \\
 5x_1 + 6x_2 - 4x_3 - x_4 = 0 \\
 3x_1 + 5x_2 - 2x_3 = 0
 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccc|c}
 1 & -3 & 1 & 2 & 0 \\
 3 & 2 & 0 & 3 & 0 \\
 5 & 6 & -4 & -1 & 0 \\
 3 & 5 & -2 & 0 & 0
 \end{array} \right) \begin{array}{l} \\ \text{II} - 3\text{I} \sim \\ \text{III} - 5\text{I} \\ \text{IV} - 3\text{I} \end{array}$$

$$\sim \left(\begin{array}{cccc|c}
 1 & -3 & 1 & 2 & 0 \\
 0 & 11 & -3 & -3 & 0 \\
 0 & 21 & -9 & -11 & 0 \\
 0 & 14 & -23 & -6 & 0
 \end{array} \right) \begin{array}{l} \\ \\ \text{II III} - 2\text{I} \\ \text{II IV} - 14\text{II} \end{array} \sim$$

$$\left(\begin{array}{cccc|c}
 1 & -3 & 1 & 2 & 0 \\
 0 & 11 & -3 & -3 & 0 \\
 0 & 0 & -36 & -58 & 0 \\
 0 & 0 & -112 + 9 & -24 & 0
 \end{array} \right) \begin{array}{l} \\ \\ \\ 36 \text{ IV} + (-112 + 9) \text{ III} \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & -3 & 1 & 2 & 0 \\ 0 & 11 & -3 & 3 & 0 \\ 0 & 0 & -36 & -58 & 0 \\ 0 & 0 & 0 & -1386 + 6382 & 0 \end{array} \right)$$

$$1) \text{ Т.к. } -1386 + 6382 = 0 \quad | : 11$$

$$582 - 126$$

$$2 = \frac{126}{58}$$

$$r(A) = r(A|B) = 3 \Rightarrow \text{с.с.}$$

$$n=4 \Rightarrow r < n \Rightarrow \text{с. неопр.}$$

$$n - r = 4 - 3 = 1 \text{ з.в. н.}$$

$$n - r = 4 - 3 = 1 - \text{св. перемен.}$$

$$\left(\begin{array}{ccc|c} 1 & -3 & 1 & -396 \\ 0 & 11 & -3 & \\ 0 & 0 & -36 & \end{array} \right) \Rightarrow x_1, x_2, x_3 - \text{з.в. пер.}$$

$$x_4 - \text{св. перемен.}$$

$$\begin{cases} x_1 - 3x_2 + x_3 + 2x_4 = 0 \\ 11x_2 - 3x_3 - 3x_4 = 0 \\ -36x_3 - 58x_4 = 0 \end{cases}$$

$$\begin{cases} x_1 = -\frac{173}{99} x_4 \\ x_2 = \frac{-47}{68} x_4 \\ x_3 = -\frac{29}{18} x_4 \end{cases}$$

$$\square x_4 = t$$

$$\text{obv. } P(-\frac{173}{99}t; -\frac{47}{66}t; -\frac{29}{18}t)$$

$$\text{LCP: } (-\frac{173}{99}t; -\frac{47}{66}t; -\frac{29}{18}t; t) =$$

$$= t(-\frac{173}{99}; -\frac{47}{66}; -\frac{29}{18}; 1) = \{(-\frac{173}{99}; -\frac{47}{66}; -\frac{29}{18}; 1)\}$$

$$2. \text{Typ } 582 - 126 \neq 0 \Rightarrow \lambda \neq \frac{126}{58}$$

$$r(A) = r(A|B) = 4 - \text{L.C.}$$

$$r = n = 4 \Rightarrow \text{L. onp.}$$

$$\begin{cases} x_1 - 3x_2 + x_3 + 2x_4 = 0 \\ 11x_2 - 3x_3 - 3x_4 = 0 \\ -36x_3 - 58x_4 = 0 \\ (638x_3 - 1386)x_4 = 0 \end{cases} \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

$$\text{O.P. } (0, 0, 0, 0)$$

$$\text{O.C.P. } \emptyset - \emptyset$$

Пример: 1) Typ $\lambda = \frac{126}{58}$. cum. cobu. onp.

$$\text{O.P. } (-\frac{173}{99}t; -\frac{47}{66}t; -\frac{29}{18}t; t); \text{O.C.P. } \{(-\frac{173}{99}; -\frac{47}{66}; -\frac{29}{18}; 1)\}$$

2) Typ $\lambda \neq \frac{126}{58}$; O.P. $(0, 0, 0, 0)$; O.C.P. $\emptyset$; cum. cobu. onp.